

CMP strain calculation

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First, we need to apply a volume conserving increase of 2% ($k = 1.02$) to the c/a ratio. Apply the constraints $\frac{c_f}{a_f} = k \cdot \frac{c_0}{a_0}$, $a_f b_f c_f = a_0 b_0 c_0$, and $\frac{b_f}{a_f} = \frac{b_0}{a_0}$ to the cell-volume equation $V = \frac{\sqrt{3}}{2} abc$, and we find that:

$$a_f = k^{-1/3} a_0, \quad b_f = k^{-1/3} b_0, \quad c_f = k^{2/3} c_0$$

Applying this to our CIF for hcp-scandium and plugging into the Bilbao strain tool yields:

$$\epsilon = \begin{pmatrix} -0.006588 & 0 & 0 \\ 0 & -0.006588 & 0 \\ 0 & 0 & 0.013288 \end{pmatrix}$$

This aligns with intuition that, when applying a strain along the c-axis, the b-a plane surface area must shrink by the inverse factor to preserve volume.